



Parameters, variables

16.2 Conduction

↳ Thermal Diffusivity

$$\alpha = \frac{k}{\rho c_p} = \frac{\text{Heat conduction}}{\text{Heat storage}}$$

17.6 Fins

↳ Fin efficiency:

$$\eta_{fin} = \dot{Q}_{fin} / \dot{Q}_{fin, max}$$

$$= \frac{\text{Actual heat transfer rate}}{\text{Ideal heat transfer rate if whole fin at base temperature}}$$

↳ Fin Effectiveness

$$\epsilon_{fin} = \frac{\dot{Q}_{fin}}{\dot{Q}_{no fin}} = \frac{\dot{Q}_{fin}}{h A_b (T_b - T_{\infty})} = \frac{\text{Rate from fin, base area } A_b}{\text{Rate from surface, area } A_b}$$

↳ Overall Fin effectiveness

$$\epsilon_{fin, overall} = \frac{\dot{Q}_{total, fin}}{\dot{Q}_{total, no fin}} = \frac{(A_{unfin} + \eta_{fin} A_{fin})}{A_{no fin}}$$

$$\dot{Q}_{fin} = \eta_{fin} \dot{Q}_{fin, max} = \eta_{fin} h A_{fin} (T_b - T_{\infty})$$

18.1 Lumped system

↳ Biot number

$$Bi = \frac{h \Delta T}{k_c \Delta T} = \frac{h L_c}{k} = \frac{\text{Convection at surface}}{\text{Conduction within body}} = \frac{\text{Internal resistance to conduction}}{\text{External resistance to convection}}$$

$Bi \text{ small} \rightarrow \text{Small temperature gradient}$

19.1 Convection

19.2 Thermal boundary layer

↳ Nusselt Number

$$Nu = \frac{h L_c}{k} = \frac{h \Delta T}{k \Delta T / L} = \frac{\text{Heat transfer by convection}}{\text{Heat transfer by conduction}}$$

↳ Prandtl Number

$$Pr = \frac{\nu}{\alpha} = \frac{\mu c_p}{k} = \frac{\text{Molecular diffusivity of momentum}}{\text{Molecular diffusivity of heat}}$$

Relative thickness of boundary layers

↳ Reynolds Number

$$Re_x = \frac{\rho V_{\infty} x}{\mu} = \frac{V_{\infty}}{\nu} \rightarrow \begin{matrix} \text{inertial stress} \\ \text{viscous stress} \end{matrix} \longrightarrow \begin{matrix} Re < 5 \times 10^5 \rightarrow \text{Laminar} \\ Re > 5 \times 10^5 \rightarrow \text{Turbulent} \end{matrix}$$

Critical value

Laminar / Turbulent flow

19.3 Parallel Flow Over Flat Plates

$$\text{Laminar, } Nu = \frac{h L}{k} = 0.664 Re^{0.5} Pr^{1/3}, \quad Re < 5 \times 10^5, \quad Pr > 0.6$$

Liquid metals, $Pr \ll 1$

$$\text{Turbulent, } Nu = \frac{h L}{k} = 0.023 Re^{0.8} Pr^{1/3}, \quad 5 \times 10^5 \leq Re \leq 10^7, \quad 0.6 \leq Pr \leq 60$$

$$\hookrightarrow Nu_x = 0.565 Re^{1/2}$$

$$Pe = Re \cdot Pr$$

$$\text{Mixed, } Nu = \frac{h L}{k} = (0.023 Re^{0.8} + 871) Pr^{1/3}, \quad 0.6 \leq Pr \leq 60, \quad 5 \times 10^5 \leq Re \leq 10^7$$

1.4 Flow across cylinders, spheres

↳ Cylinders

$$Nu = \frac{hD}{k} = C Re^m Pr^{\frac{1}{4}} \quad \text{Cons.} \quad @ \text{ Film temperature}$$

↳ Sphere

$$Nu = \frac{hD}{k} = 2 + [0.4 Re^{\frac{1}{4}} + 0.06 Re^{\frac{1}{2}}] Pr^{\frac{1}{4}} \left(\frac{\mu_{\infty}}{\mu_s} \right)^{\frac{1}{4}} \quad @ T_{\infty} \quad (\text{except } \mu_s @ T_s)$$

2.1.3 Blackbody Radiation

↳ Blackbody emissive power (Energy/time, unit area), given by Stefan Boltzmann Law

$$E_b(T) = \sigma T^4 \quad [W/m^2]$$

↳ Planck's Law

$$E_{b\lambda}(\lambda, T) = \frac{C_1}{\lambda^5 (\exp(C_2/\lambda T) - 1)} \quad \text{energy radiated by blackbody at temperature } T / \text{unit time, surface area, unit wavelength at } T, \lambda$$

↳ Wien's Displacement Law

$$(\lambda T)_{\text{max power}} = 2897.8 \mu m \cdot K$$

↳ Total emissive Power

$$E_b(T) = \int_0^{\infty} E_{b\lambda}(\lambda, T) d\lambda = \sigma T^4$$

Planck's Law Stefan Boltzmann Law

↳ Blackbody radiation function (dim-less)

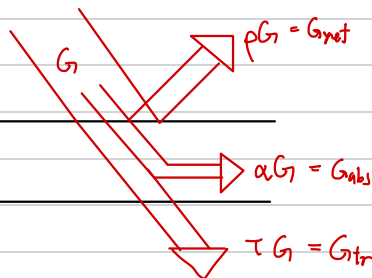
$$f_{\lambda}(T) = \frac{\int_0^{\lambda} E_{b\lambda}(\lambda, T) d\lambda}{\sigma T^4}$$

$$f_{\lambda_1-\lambda_2}(T) = \frac{\int_{\lambda_1}^{\lambda_2} E_{b\lambda}(\lambda, T) d\lambda}{\sigma T^4} = f_{\lambda_2}(T) - f_{\lambda_1}(T)$$

2.1.4 Radiative Properties

↳ Hemispherical emissivity

$$\epsilon_{\lambda}(\lambda, T) = E_{\lambda}(\lambda, T) / E_{b\lambda}(\lambda, T)$$



↳ Gray → Properties indep of λ

↳ Diffuse → Properties indep of direction

Diffuse Reflection → rough surface



Specular reflection → smooth surface



↳ Absorptivity

$$\alpha = \frac{G_{abs}}{G}$$

↳ Reflectivity

$$\rho = \frac{G_{refl}}{G}$$

↳ Transmissivity

$$\tau = \frac{G_{tr}}{G}$$

$$G = G_{abs} + G_{tr} + G_{refl}$$

$$1 = \alpha + \rho + \tau$$

↳ Black body → $\alpha=1, \rho=0, \tau=0$

↳ Opaque surface

solid and liquid → $\tau=0, \alpha+\rho=1$

↳ Gases → $\rho=0, \alpha+\tau=1$

Kirchoff's Law

$\epsilon = \alpha$ for same surface temperature
and temperature of source radiation

21.5 View Factor

F_{ij} or $F_{i \rightarrow j}$, view factor

$F_{ij} = 0$ den radiation from i cannot strike j

$F_{ij} = 1$ when all radiation from i is intercepted by j

F_{ii} , self view factor

$F_{ii} = 0$ for convex or plane surface

$F_{ii} \neq 0$ for concave surface

21.6 View factor relation

$\hookrightarrow$ Reciprocity

$$A_i F_{ij} = A_j F_{ji}$$

$F_{ij} = F_{ji}$ when $A_i = A_j$

$\hookrightarrow$ Summation

$$\sum_{j=1}^N F_{ij} = 1$$

$\hookrightarrow$ Result of summation

Evaluated view factors for N surface:

$$\frac{1}{N(N-1)}$$

$\hookrightarrow$ Superposition

$$F_{1(2,3)} \text{ or } F_{1 \rightarrow (2,3)} = F_{1 \rightarrow 2} + F_{1 \rightarrow 3}$$

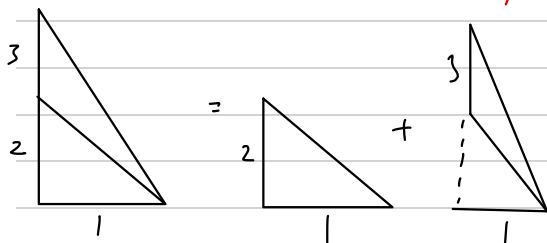
But $F_{(2,3) \rightarrow 1} \neq F_{2 \rightarrow 1} + F_{3 \rightarrow 1}$

$$A_1 F_{1 \rightarrow (2,3)} = A_1 F_{1 \rightarrow 2} + A_1 F_{1 \rightarrow 3}$$

$\hookrightarrow$ Symmetry

if j, k are symmetric about i , then $F_{ij} = F_{ik}$, $F_{ji} = F_{ki}$

$$(A_2 + A_3) F_{(2,3) \rightarrow 1} = A_2 F_{2 \rightarrow 1} + A_3 F_{3 \rightarrow 1} \quad \text{reciprocity}$$



21.7 Blackbody Radiation

$$\dot{Q}_{1 \rightarrow 2} = A_1 E_{b1} F_{1 \rightarrow 2} - A_2 E_{b2} F_{2 \rightarrow 1}$$

$$= A_1 F_{1 \rightarrow 2} \sigma (T_1^4 - T_2^4)$$

21.8 Radiation heat transfer, diffuse, gray surfaces

$\hookrightarrow$ Radiosity, total radiation energy leaving a surface
per unit time, unit area

J_i = Radiation emitted + radiation reflected

$$= \epsilon_i E_{bi} + \rho_i G_i$$

For gray, opaque surfaces, $\alpha_i + \rho_i = 1$, $\epsilon_i = \alpha_i$

$$= \epsilon_i E_{bi} + (1 - \epsilon_i) G_i$$

($= E_{bi}$ for black bodies)

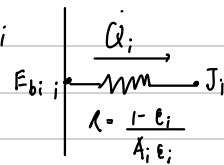
$\hookrightarrow$ net rate of radiation heat transfer

$\dot{Q}_i$ = Radiation leaving i - Radiation incident on i

$$= A_i (J_i - G_i)$$

$$= \frac{A_i \epsilon_i}{1 - \epsilon_i} (E_{bi} - J_i)$$

$$= \frac{E_{bi} - J_i}{R_i}, \quad R_i = \frac{1 - \epsilon_i}{A_i \epsilon_i}, \quad \text{Invalid for blackbodies } R_i = 0$$



$\hookrightarrow$ Re-radiating surface

$$J_i = E_{bi} = \sigma T_i^4, \quad \dot{Q}_i = 0$$

↳ Radiation b/w **any** 2 surfaces

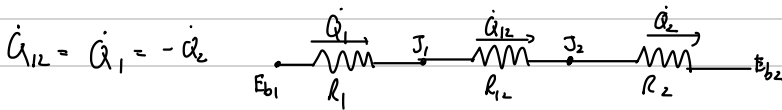
$$\begin{aligned}\dot{Q}_{i \rightarrow j} &= \text{radiation leaving surface } i \text{ striking surface } j - \text{radiation leaving surface } j \text{ striking surface } i \\ &= F_{i \rightarrow j} A_i J_i - F_{j \rightarrow i} A_j J_j \\ &= F_{i \rightarrow j} A_i (J_i - J_j) \\ &= \frac{(J_i - J_j)}{R_{i \rightarrow j}}, \quad R_{i \rightarrow j} = \frac{1}{F_{i \rightarrow j} A_i} \quad R_{i \rightarrow j} = R_{j \rightarrow i} \text{ by reciprocity}\end{aligned}$$

↳ Radiation b/w **n** surfaces

$$\dot{Q}_i = \sum_{j=1}^n \dot{Q}_{i \rightarrow j} = \sum_{j=1}^n A_i F_{i \rightarrow j} (J_i - J_j) = \sum_{j=1}^n \frac{J_i - J_j}{R_{i \rightarrow j}}, \quad \dot{Q}_{i \rightarrow i} = 0 \text{ for any shape.}$$

$$\frac{E_{bi} - J_i}{R_i} = \sum_{j=1}^n \frac{J_i - J_j}{R_{i \rightarrow j}}$$

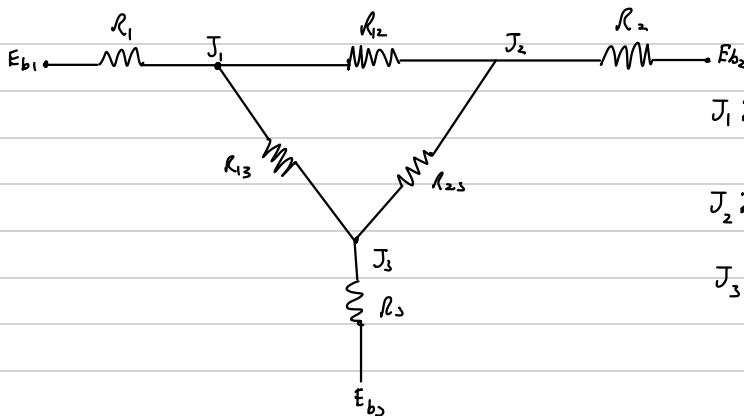
↳ Radiation b/w 2 enclosed surfaces



$$\dot{Q}_{12} = \frac{E_{b1} - E_{b2}}{R_1 + R_{12} + R_2} = (E_{b1} - E_{b2}) \left(\frac{1}{\frac{1-\epsilon_1}{A_1 \epsilon_1} + \frac{1}{A_1 F_{12}} + \frac{1-\epsilon_2}{A_2 \epsilon_2}} \right)$$

$$= \sigma (T_1^4 - T_2^4) \left(\frac{1}{\frac{1-\epsilon_1}{A_1 \epsilon_1} + \frac{1}{A_1 F_{12}} + \frac{1-\epsilon_2}{A_2 \epsilon_2}} \right)$$

↳ 3 enclosed surfaces



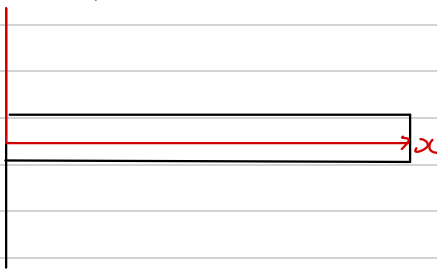
"Kirchhoff's Current Law"

$$J_1: \frac{E_{b1} - J_1}{R_1} + \frac{J_2 - J_1}{R_{12}} + \frac{J_3 - J_1}{R_{13}} = 0$$

$$J_2: \frac{E_{b2} - J_2}{R_2} + \frac{J_1 - J_2}{R_{12}} + \frac{J_3 - J_2}{R_{23}} = 0$$

$$J_3: \frac{E_{b3} - J_3}{R_3} + \frac{J_2 - J_3}{R_{23}} + \frac{J_1 - J_3}{R_{13}} = 0$$

Fin equation



Heat conduction rate at x = Heat conduction rate at $x + \Delta x$ + Convection

$$\dot{Q}_{\text{cond},x} = \dot{Q}_{\text{cond},x+\Delta x} + \dot{Q}_{\text{conv}}$$

$$\dot{Q}_{\text{conv}} = h(p \Delta x)(T - T_{\infty})$$

$$\dot{Q}_{x+\Delta x} - \dot{Q}_x + h(p \Delta x)(T - T_{\infty}) = 0$$

$$\frac{d\dot{Q}_{\text{cond}}}{dx} + hp(T - T_{\infty}) = 0$$

$$\dot{Q}_{\text{cond}} = -kA \frac{dT}{dx}$$

$$-kA \frac{d^2T}{dx^2} + hp(T - T_{\infty}) = 0$$

$$\text{Let } \theta = T - T_{\infty}$$

$$-kA \frac{d^2\theta}{dx^2} + hp\theta = 0$$

$$\frac{d^2\theta}{dx^2} - \frac{hp}{kA}\theta = 0, \quad \alpha^2 = \frac{hp}{kA}$$

$$\frac{d^2\theta}{dx^2} = \alpha^2\theta$$

$$\theta = C_1 e^{\alpha x} + C_2 e^{-\alpha x}, \quad \text{Boundary cond. } \theta(0) = T_b - T_{\infty}$$

1.) For infinitely long fin,

$$a) \quad x \rightarrow \infty, \quad \theta = 0$$

$$0 = C_1 e^{\alpha \infty}$$

$$C_1 = 0$$

$$\rightarrow \frac{T - T_{\infty}}{T_b - T_{\infty}} = e^{-\alpha x}$$

$$b) \quad \theta(0) = T_b - T_{\infty}$$

$$T_b - T_{\infty} = C_2$$

$$\dot{Q} = -kA \left. \frac{dT}{dx} \right|_{x=0}$$

$$= (T_b - T_{\infty}) \alpha (kA)$$

$$= \sqrt{hp kA} (T_b - T_{\infty})$$

2.) Negligible heat loss from the fin tip

$$a) \quad \left. \frac{d\theta}{dx} \right|_{x=L} = 0$$

$$b) \quad \theta(0) = T_b - T_{\infty}$$

$$\theta = C_1 e^{\alpha x} + C_2 e^{-\alpha x}$$

$$0 = \alpha C_1 e^{\alpha L} - \alpha C_2 e^{-\alpha L}$$

$$T_b - T_{\infty} = C_1 + C_2$$

$$0 = C_1 e^{\alpha L} - C_2 e^{-\alpha L}$$

$$C_1 = \frac{T_b - T_{\infty}}{1 + e^{2\alpha L}}$$

$$C_2 = \frac{T_b - T_{\infty}}{1 + e^{-2\alpha L}}$$

$$\frac{T(x) - T_{\infty}}{T_b - T_{\infty}} = \frac{\cosh m(L-x)}{\cosh mL}$$

Lumped System

Energy balance:

Heat into body in dt = Energy increase in dt

$$hA_s (T_\infty - T) dt = mc_p dT$$

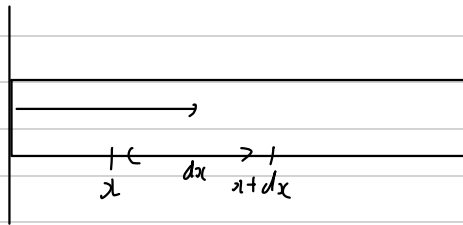
$$\frac{hA}{mc} (T_\infty - T) = \frac{dT}{dt}$$

let $T - T_\infty$ be θ

$$\frac{d\theta}{dt} = -\frac{hA}{mc} \theta$$

$$\theta = \theta(0) e^{-\frac{hA}{mc} t}, \quad \theta(0) = T_i - T_\infty$$

$$\frac{T - T_\infty}{T_i - T_\infty} = e^{-bt}$$



$$\dot{Q}_x = \dot{Q}_{x+dx} + \dot{Q}_{conv}$$

$$\dot{Q}_{conv} = hA(T - T_{\infty})$$

$$-h_p dx(T - T_{\infty}) = \dot{Q}_{x+dx} - \dot{Q}_x$$

$$-h_p(T - T_{\infty}) = \frac{d\dot{Q}}{dx}$$

$$h_p(T - T_{\infty}) = hA \frac{dT}{dx} \quad \dot{Q} = -hA \frac{dT}{dx}$$

$$\frac{d^2\theta}{dx^2} = \frac{h_p}{kA} \theta$$

$$\dot{Q} = \frac{T(x) - T_{\infty}}{T_b - T_{\infty}} \exp(-mx)$$